Screening OPERA for hourly extremes

Rules, how much each removes, what the reference data say

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14 October 2026

The setting

- **Target:** the 1-h accumulation at 2 km, OPERA's own ACRR convention: hour $H = \frac{1}{4} \times$(frames $H-45$, $H-30$, $H-15$, H).
- **Screening runs on the 15-min frames**: decide over the day, repair per frame, then sum. Tile-hours with clear errors are rejected when the splits are made.
- **Main concern:** unphysical values contaminating the extremes. Exact agreement of rain amounts with other products is secondary.
- **Ground rules**
  - a rule is kept only if independent data support it (RADKLIM, DE/CH gauges);
  - every pass criterion is written down and committed *before* the test runs;
  - no decision rests on looking at images.
- **Data used here:** 199 calibration days (100 random, 79 event, 20 known failures); RADKLIM on 193 of them; gauges on 246 days.

The rule set (current chain)

step	rule	how it is supported
frame	drizzle < 0.1 mm/h $\rightarrow 0$	noise floor
frame	static clutter (hot pixels, repair)	RADKLIM: 100% of removals dry (89 cases)
frame	spikes	signature
frame	ceiling values (repair; ≥ 5 px rejects the tile-hour)	context: 86% not inside intense rain
frame	footprints of > 500 mm/h cores	physical bound
frame	range rings ≥ 89 mm/h	climatological geometry
frames	RLAN rays (site-anchored, fixed over ± 4 frames)	signature
day	local-peak persistence ($p_1 \geq 24$ of 96 frames)	lowering it fails RADKLIM
frame	cells without temporal support	RADKLIM: 97–100% dry
hour	isolated single-frame bursts	RADKLIM: 87.7% dry
split	radar-frame failures, repeated values, refused repairs	signature
split	radar-day exclusion (score ≥ 1)	RADKLIM: 99.8% dry; rest of day worse

How much each rule removes: random days (100)

Share of pixel-hours, by *raw* hourly total, that the rule lowers by $\geq 50\%$. Rows "rejected" = pixel-hours dropped because their tile-hour is rejected.

rule	5–10 mm	10–20	20–50	50–100	≥ 100 mm
static clutter	1.15%	4.31%	**13.0%**	**25.8%**	**36.5%**
spikes	0.60	1.22	2.17	3.51	6.57
ceiling repair	–	–	–	14.5	4.48
> 500 mm/h footprints	–	–	1.04	6.09	**44.3**
range rings	–	–	0.11	8.76	1.89
persistence	1.88	2.69	2.43	2.03	0.69
no temporal support	0.22	0.38	0.91	0.94	0.18
bursts	–	–	5.59	4.62	0.52
rejected: ceiling tile	0.02	0.09	0.91	3.29	0.18
rejected: radar-day	1.11	1.48	1.42	0.99	0.14
left intact	95.0%	89.6%	71.9%	29.0%	4.2%

Below 5 mm: 98.5% left intact. Hourly totals ≥ 100 mm are rare in reality, so a 4% survival there is expected.

How much each rule removes: event days (79)

Same measure on the days of the catalogued storms.

rule	5–10 mm	10–20	20–50	50–100	$\geq$ 100 mm
static clutter	0.63%	2.21%	8.25%	**21.5%**	35.5%
spikes	0.31	0.61	1.32	3.07	3.26
ceiling repair	–	–	–	7.03	6.02
$>$ 500 mm/h footprints	–	–	0.47	6.00	24.0
range rings	–	–	0.07	8.62	2.50
persistence	0.91	1.49	2.62	4.32	1.41
no temporal support	0.16	0.23	1.38	**6.09**	0.41
bursts	–	–	2.67	4.72	0.68
rejected: unphysical	0.01	0.08	0.87	0.74	19.0
rejected: ceiling tile	0.02	0.07	0.67	6.43	0.66
rejected: radar-day	0.89	1.22	1.54	1.25	0.17
left intact	97.1%	94.0%	79.7%	30.3%	6.3%

Event days contain artefacts too: these are not real-rain losses as such. The tests on the next slide decide which are.

Is what each rule removes really an artefact?

RADKLIM presence test: a removal is "dry" when RADKLIM has < 5 mm within ± 2 px and ± 1 h. Bar: $\geq 85\%$ dry. The disagreement floor of ordinary values is 5–6%.

test	evidence	result	decision
static-clutter repair	89 removals, 100% dry	**pass**	stays
no temporal support	277 removals, 97–100% dry	**pass**	stays
burst rule	308 removals at 20–50 mm, 87.7% dry	**pass**	adopted
isolated ≥ 31 mm/h pixels (radar-day)	69,726 pixel-frames, 99.8% dry	**pass**	stays
rest of an excluded radar-day	false alarms 40% vs 12% (control)	worse	exclude whole day
ceiling pixels	13.7% inside intense rain	artefacts	stays
persistence p_1 24 $\rightarrow$ 20 or 16	66–80% dry, 24–59 confirmed real	**fail**	stays 24
Mahalanobis outlier mask	84% / 76% dry	**fail**	not adopted
radar-year exclusion	removed most event data	–	dropped

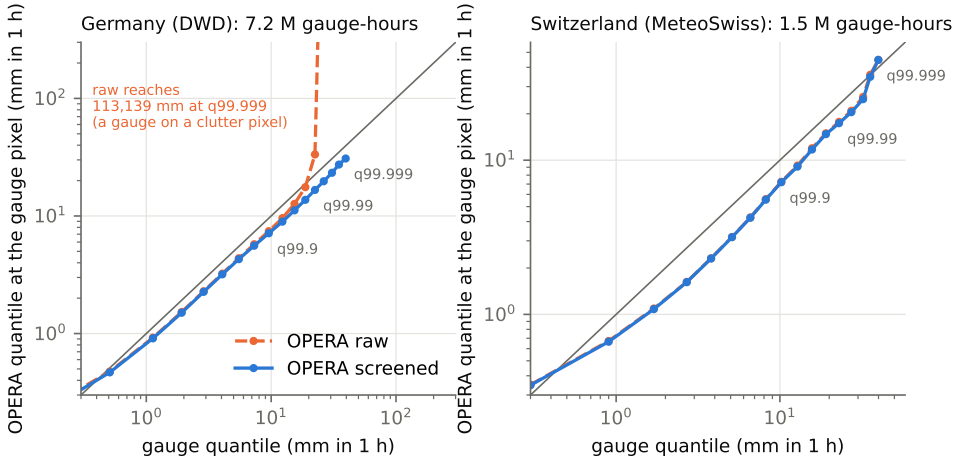
RADKLIM is reprocessed from the same German radars, so it can confirm an artefact both share. The tests cover Germany only; elsewhere rules rest on their signature.

Readiness gate before the full-archive build

criterion	result	status
1. known failures (Madrid, Valjevo, 48.62) kept	0 / 0 / 0 tile-hours	**pass**
2a. false alarms vs RADKLIM at 20 mm, same gauge-hours	OPERA 6.3%, RADKLIM 0.9%: +5.4 points [+2.3, +9.9]; bar $\leq +2$	**fail**
2b. heavy gauge-hours the screen removes	0.1%	**pass**
3a. event tile-hours rejected	0% except on excluded radar-days	**pass**
3b. rejection rising with intensity	ceiling, radar-day: artefact-driven	explained
3 / 5. tail stability, $\xi(u)$ drift	need the built test split	after build

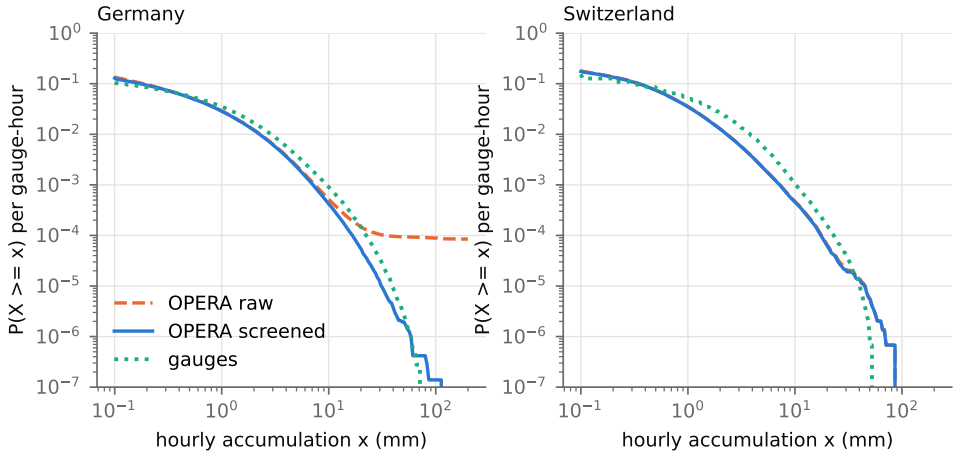
The gate turns on 2a. Of the 24 screened false-alarm gauge-hours, about 9 are clear artefacts (RADKLIM ≈ 0). The largest: a stationary echo at Kleiner Feldberg on 2018-04-30, 6 hours of 23–83 mm in a dry ring, a local peak in 20 frames (the threshold is 24). About 15 sit inside convection (RADKLIM 5–20 mm): most likely the gauge missed a small cell.

Distributions against gauges (point reference)



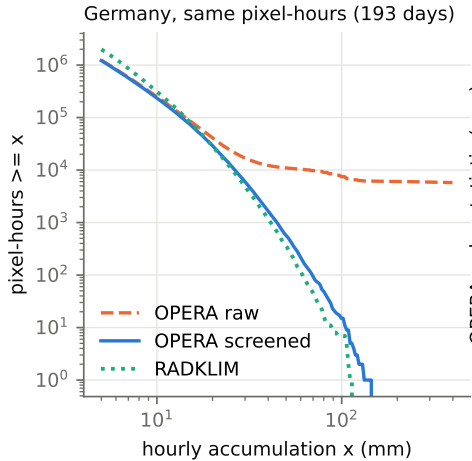
Quantiles of OPERA at the gauge pixel against the gauge, same gauge-hours (246 days). Raw OPERA in Germany is dominated above q99.99 by a gauge on a clutter pixel; screened OPERA sits at 0.71–0.78× the gauge quantile from q99.9 to q99.999 (Switzerland 1.12× at q99.999, few cases): about 25% low, from point vs 2 km area and radar underestimation.

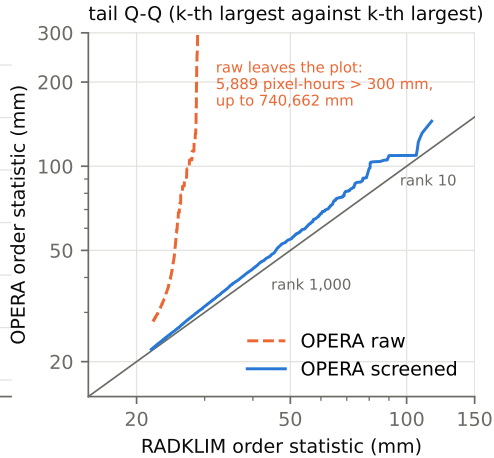
Exceedance against gauges



$P(X \geq x)$ per gauge-hour. In Germany the screen removes a plateau of spurious values at $\sim 10^{-4}$; afterwards OPERA sits slightly below the gauges at every level.

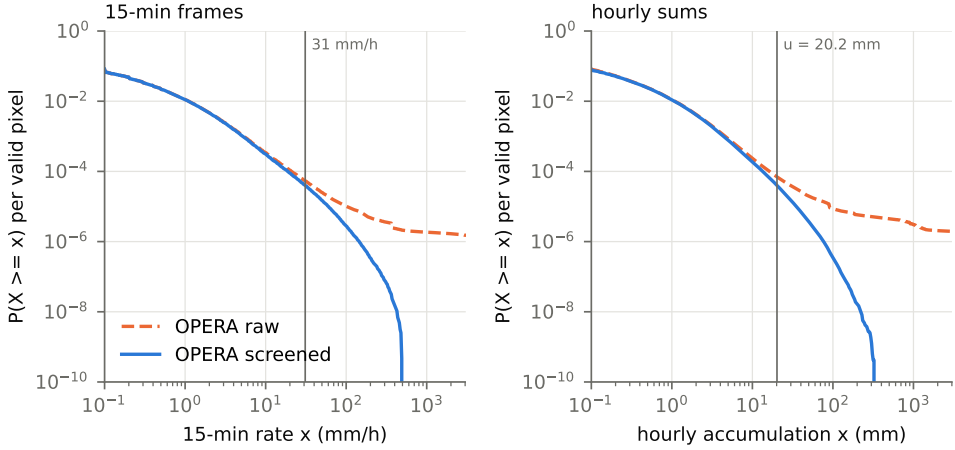
Against RADKLIM on the same German pixel-hours





Screened OPERA matches RADKLIM at 20 mm and has 1.2× / 1.6× / 2.5× its counts at 30 / 50 / 75 mm. RADKLIM underestimates peaks, so this excess bounds the residual contamination from above; it does not measure it.

Europe: raw against screened (100 random days, full grid)



$P(X \geq x)$ per valid pixel. The screen leaves everything below ~ 10 mm/h (or mm) untouched and cuts the spurious plateau of the raw tail, 10^{-6} to 10^{-5} up to 10^{3}; the screened 15-min tail ends at the 500 mm/h physical bound. **Provisional hourly tail threshold:** 31 mm/h is exceeded by 4.0×10^{-5} of screened 15-min pixel-frames; the hourly sums exceed $u = \mathbf{20.2\ mm}$ at the same frequency (final u from a stability analysis on the build).

Decisions this week

- Radar ownership from the radars listed in each composite (Italy never contributes; Austria never; CH and HU from 2014).
- Radar-day exclusion by a pre-stated score (≥ 1, one decade above both the radar's own seasonal q99 and its neighbours); **radar-year rule dropped**.
- Static-clutter repair and the temporal-support rule **kept**; persistence threshold **kept at 24**.
- `min_wet_jump` 0.15 $\rightarrow$ 0.021, now derived (q99.9 of normal operation); same 48 flagged frames.
- Burst rule adopted; Mahalanobis mask not adopted.

Limitations to state

- Independent data cover Germany and Switzerland only; elsewhere rules rest on their signature (Valjevo, Madrid and Oradea lie outside).
- RADKLIM shares the German radars and checks presence, not magnitude.
- The screen targets overestimation: blockage, attenuation and overshoot (Alps) are not corrected, so the tail is biased low there.
- Up to about a third of kept values ≥ 75 mm over Germany are RADKLIM-dry.
- The plain hourly sum beads fast cells; the advection-corrected field is stored alongside.
- Criteria were fixed in advance internally (commit timestamps), not registered externally.

Next steps

1. A rule for single-day stationary echoes (fixed location, dry ring, many hours): the remaining cause of the 2a failure. Test it like the others.
2. Full-archive hourly scan ($\sim$ 30 h), splits and store.
3. Gate criteria 3 and 5 on the build; hourly tail threshold from a stability analysis.
4. Retrain the backbones on v4 and recompute σ_r.